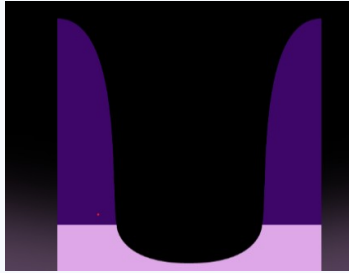
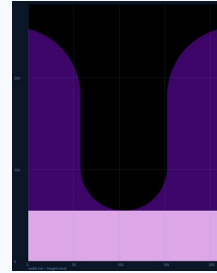


Building a U-shaped mask profile

Reproduce a flared mask with a rounded (U) opening bottom using the material stack and per-layer shape treatments.



Target — your image



Result in Profile Studio

Steps

1. **Material stack** (row 1 = top). Add **indigo** 200 nm, then **lavender** 55 nm.
2. **Base feature**. Pitch **210**, Space **95**, Opening depth **200**, Top vacuum **25**.
3. **Opening bottom round**: **47** nm (about half of Space) — this makes the U-shaped bottom.
4. **Shape the mask**. On the indigo layer press its **shape** button and add **Round** radius **75** for the flared top.
5. The preview updates live as you edit — click **Save .bmp...** to export the image.

How the shape treatments work

- **Round (radius)** — a fillet: a quarter-circle taken off the top corner.
- **Chamfer (size) / Facet (angle + depth)** — straight-cut corners. Chamfer is a fixed **45°** clip; facet does the same at any angle (depth = how far down the wall, angle measured from the horizontal top). A 45° facet is exactly a chamfer.
- **Taper (angle)** — slopes the **whole** layer wall, not just the corner. Its angle is measured from the horizontal **base**: **90° = vertical**, smaller = more sloped.
- A treatment larger than its layer is **clamped to that layer**, so it eases into the wall instead of leaving a step.
- You can **stack several treatments** on one layer (e.g. Round + Taper) via **+ Add another treatment** — they combine and are order-independent.

Working in the preview

- **Zoom** — scroll the wheel over the preview (or use the Zoom slider). Double-click or **Reset** returns to the full view.
- **Move** — once zoomed in, drag (or use the arrow keys) to reposition. At 1× the whole profile is already in view, so there is nothing to move.
- **Measure** — drag a line across any feature to read it in nm. Set **Snap** to Edge/Vertex to lock endpoints onto material boundaries. Switch back to **Move** to pan.
- **Smoothing** (Off / 2× / 4× / 8×) de-jags curved walls without ever blending colours — the .bmp keeps exactly one colour per material.
- **Save .bmp...** exports the picture; **Polygons...** exports the profile as editable vector shapes (SVG) plus exact nm coordinates (JSON).

Tips

- The version pill (top-right) and title bar show your build — quote it when reporting an issue.
- The window starts maximised and is resizable; the preview grows with it. The plot grid itself stays a fixed size — only the axis numbers change.
- Toolbar Save stores the whole project; Save .bmp exports just the picture. Add your own colours with “+ New material (color)”.